

PAC-MAN (1980)

Complete Restoration Guide for Public Placements

Cabinet • Controls • Power • Monitor • Reliability

Prepared for **Coin Op 4 Charity** / Arcade Angels

Making a classic machine reliable and inviting for youth centers and community spaces

1. Overview & Goals

Pac-Man (Namco / Midway, 1980) remains one of the strongest draws for all ages. For public and youth-center placements the priorities are reliability, safety, clean presentation, and a smooth player experience. This guide covers the whole machine so a restored unit can be placed with confidence and stay in service with minimal call-backs.

2. Safety

- Always unplug the cabinet before opening it.
- CRT monitors store high voltage. Discharge the anode properly before working on the chassis or neckboard.
- Original linear power supplies and isolation transformers involve AC voltages — treat them with respect.
- If you are not comfortable with CRT work, an LCD conversion is a practical and safer path for high-use public sites.

3. Cabinet Exterior & Cosmetics

- Clean the entire cabinet thoroughly. Remove sticky residue, dust, and decades of hand prints.
- Inspect side art, front decals, bezel, and control-panel overlay. Replace or carefully touch up heavily damaged graphics so the machine looks cared for.
- Marquee: clean the plastic, replace the fluorescent tube and starter if dim or flickering. A bright marquee makes the game far more inviting.
- Coin door: clean, lubricate hinges, confirm locks work or replace for the placement site. Set free-play or appropriate pricing and document it.
- Level the cabinet so it sits solidly with no rocking.

4. Control Panel, Joystick & Buttons

The control panel is the player's main contact with the machine. Clean or replace sticky buttons. Inspect the leaf switches or microswitches under the joystick and buttons; clean contacts and adjust for positive, reliable actuation. A worn or loose joystick can be rebuilt with new bushings, actuators, and a fresh restrictor plate if needed. If the control-panel overlay is heavily scratched, torn, or peeling, a replacement improves both appearance and hygiene for public use. Secure any lifting edges.

5. Power Supply & Fuses

Original Pac-Man cabinets use a classic linear power supply with a large transformer and an isolation transformer for the monitor. These supplies are often filthy and the fuse holders become oxidized or burned. Clean the supply thoroughly. Check and replace fuse holders if the clips are weak or scorched. Verify correct fuse values and that AC voltages reach the board edge connector (commonly ~7 VAC and ~12 VAC depending on exact wiring). Many operators convert to a modern switching power supply for cleaner, more stable power and fewer edge-connector burn issues. Either path is acceptable; the goal is solid, stable voltage at the PCB under load.

6. Game PCB & Edge Connector

Clean the edge connector fingers on the PCB with a proper contact cleaner or pencil eraser (gently) and isopropyl alcohol. Inspect the cabinet-side connector for burned or loose pins — a common failure point that causes intermittent boots, resets, or missing voltages. Reseat the board firmly. If the machine has a history of random resets or “dead” behavior with power present, the edge connector and power path are the first places to look before assuming a bad board.

7. Monitor

CRT path (original G07 / Wells Gardner style)

Common issues include weak or missing colors, focus problems, brightness/contrast drift, and chassis failures (caps, flyback, etc.). Start with cleaning the chassis, checking fuses on the monitor, confirming isolation-transformer power, and reseating the neckboard. Cap kits and known service procedures exist for the popular chassis used in Pac-Man. Only attempt deeper CRT work if you are experienced and follow proper discharge procedures.

LCD conversion (practical for public placements)

An LCD conversion reduces heat, eliminates high-voltage hazards, and provides consistent picture quality with lower ongoing maintenance. Kits and mounting solutions exist; some cabinet modification or shimming is often required. Preserve the original CRT and chassis labeled and stored when possible — they remain useful donor parts or can be restored later.

8. Sound, Coin Door & Miscellaneous

- Confirm the speaker works cleanly at reasonable volume. Replace if the cone is torn or the sound is distorted.
- Coin door switches and lockout coils (if still used) should operate cleanly. For free-play youth placements, configure and document free-play settings.
- Clear dust from the interior, especially around the power supply and any vents. Heat shortens component life.
- Check the interlock switch (rear door) so it functions correctly for safety.

9. Final Testing Checklist Before Placement

- Full cold boot to attract mode with correct colors and sound
- Joystick moves cleanly in all four directions with no dead zones or sticking
- All buttons (including Start and coin switches) register reliably
- Complete several full games with no freezes, resets, or odd behavior
- Monitor picture is focused, bright enough, and free of major geometry issues from normal player distance
- Cabinet exterior is clean, level, and free of sharp edges or loose parts
- Marquee is lit and readable
- Free-play or pricing is configured correctly for the host site
- Keys, any spare fuses, and basic notes are bagged and labeled for the host organization

10. Notes for Host Sites

Provide a short one-page sheet: how to power the machine on/off, free-play status, and who to contact for problems. Recommend a simple weekly wipe-down of the control panel and glass. Ask the host to report any new stickiness, picture changes, or resets early. Document any non-original parts (LCD, switching supply, new overlay, etc.) so future volunteers know what they are looking at.

This guide is intended for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. The goal is a Pac-Man that stays reliable, safe, and fun for the people who actually play it.

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